XAVIER CHEN

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RESEARCH INTERESTS

Hierarchical LLM-based agent systems (multimodal; multi-agent) and long-horizon multi-turn dynamics under prompting and memory strategies; neuro-inspired memory and control loops; cognitively grounded evaluation of LLM reasoning vs. human reasoning (cognitive modeling, metacognition/self-monitoring); high-performance ML systems (GPU optimization, parallel/distributed computing, cloud infrastructure).

EDUCATION

University of California, San Diego, La Jolla, United States — Expected Jun 2026

Master of Science in Computer Science

- GPA: 3.925/4.0 (MS CS)
- Neuroscience coursework: 4.0/4.0 (recorded on a separate transcript per department policy)
- Selected Coursework: Deep Learning, Systems Neuroscience, Parallel Computation

Tongji University, Shanghai, China — Aug 2018 - Jul 2023

Bachelor of Engineering in Vehicle Engineering (Automobile), Minor in Artificial Intelligence

- GPA: 4.34/5.00. Equal to 3.84/4.00 in US system

MANUSCRIPTS & TECHNICAL REPORTS

1. **Xavier Chen.** "**Human-Inspired Preprocessing for Robust Face Recognition.**" *Research technical report*, Cottrell Lab, UC San Diego, 2025 **[PDF] [CODE]**
2. **Xavier Chen***, Grace Moreau, Brian Nguyen, Nikita Kachappilly, Ben Bao. "**Forgetful Memory Machines: Hopfield Networks and How to Improve Them.**" *Course final project report*, **CSE 251B (Deep Learning)**, Instructor: Prof. Garrison Cottrell, UC San Diego, 2025 **[PDF] [CODE]**
3. Yinsheng Wu, Yiyue Wang, **Xavier Chen**, Xiangyu Feng. "**Optimized Design of Bus Rapid Service Express Stop Patterns.**" *Student Innovation Training Program report (funded)*, Tongji University, 2020. **[PDF]**

* denotes corresponding/lead author.

RESEARCH EXPERIENCE

Cottrell Lab, UC San Diego — Researcher — Feb 2025 – Mar 2026

Advisor: Prof. Garrison Cottrell

- Studied how biologically inspired transforms (foveation, log-polar mapping) affect learned representations, using the face inversion effect as a probe of model–human differences.
- Implemented fully connected and convolutional attractor-style recurrent blocks; adding them drove inverted-face accuracy toward chance, suggesting stronger latent constraints under inversion/representation perturbation.
- Developed a saccade-inspired salience-map augmentation module; improved inverted-face accuracy to 55% vs. 20% for a vanilla CNN baseline.
- Manuscript in preparation for **CogSci 2026**. (planned submission Feb 2026)

Tongji University — Research Assistant — Apr 2023 – Jul 2023

Advisor: Prof. Lianghua He

- Designed and ran a cognitive EEG experiment (64-channel cap) to test how contextual stimuli (images and native-language descriptions) modulate neural responses across four scenario categories.
- Built preprocessing pipeline in MNE-Python and organized labels for supervised learning.
- Implemented/adapted an EEG-Net-based classifier; achieved ~40% 4-class accuracy (chance 25%) on {person, fruit, sport, street}.

Technical University of Munich — Researcher Apr 2022 – Sep 2022
Advisor: Prof. Daniel Renjewski

- Built MATLAB/Simulink models for a bipedal robot, progressing from mass–spring abstractions to a 6-DoF dynamics model; designed a 20-neuron fixed-weight neural controller with comparable locomotion behavior.
- Optimized controller parameters using Particle Swarm Optimization, achieving stable walking at 1.5 m/s in MATLAB simulation.

Optimization Design for Express Bus Stops Mar 2019 – Sep 2020
Student Innovation Training Program (Funded), Tongji University, Advisor: Prof. Jing Teng

- Led a funded project on express-stop design motivated by gaps between scheduled and realized bus wait times.
- Collected survey and observation data (stop spacing, boarding/leaving patterns) across multiple rides.
- Fit an MLE discrete-choice model (regular vs. express vs. walking) using walking/waiting/in-vehicle time features; used it to propose improved express-stop patterns.
- Produced an order-of-magnitude cost/time savings estimate under documented assumptions; summarized results in a program report.

TEACHING & MENTORING

Teaching Assistant, UC San Diego, Department of Cognitive Science

- DSGN 1 - Design of Everyday Things (Mar – Jun 2025)
- COGS 17 - Neurobiology of Cognition (Sep – Dec 2025)
- COGS 107B - Systems Neuroscience (Jan – Mar 2026)

Responsibilities: led discussion sections; prepared materials; supported grading and student support.

SELECTED PROJECT

Parallel Computation Project Sep 2025 – Dec 2025
Code: private repo (available upon request).

- Accelerated single-core AArch64 GEMM using packing and SIMD; improved from 6 to 23.5 GFLOP/s, exceeding OpenBLAS (19 GFLOP/s) on the same hardware.
- Optimized CUDA GEMM on NVIDIA Tesla T4 using shared-memory blocking + outer-product microkernel + 2D interleaved tiling; reached 3.4 TFLOP/s (FP32) (~44% of theoretical peak).
- Implemented an MPI-based parallel 5-point stencil solver for the Aliev–Panfilov cardiac model.

AWARDS

- Shell Eco-marathon China: 3rd place (Z201), 4th place (Z202), 2021
- Tongji Outstanding Student Scholarship: First-Class (2019), Third-Class (2020–2021)

SKILLS

- **Languages:** Python, C/C++, CUDA, MATLAB, Java
- **ML/Scientific:** PyTorch, NumPy, LangChain, Scikit-learn
- **Systems:** Linux, Git, Docker, Kubernetes, GCP, MPI, SIMD/NEON

CERTIFICATIONS

MIT xPRO, Professional Certificate in Data Engineering Jul 2024
Google Cloud, Professional Machine Learning Engineer Certification Aug 2025 – Aug 2027